Vansh Agrawal

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B.Tech, Computer Science & Engineering (AI & ML) | SRM IST, Kattankulathur | Graduating 2027

Summary

Final-year CS student and AI/ML Engineer with production experience in **computer vision**, **multi-agent RAG systems**, and **real-time edge intelligence**. Currently driving automated quality control at **Tata Steel** using OpenCV, achieving **95% inspection precision**. Proven track record delivering **low-latency, deployable ML pipelines** across autonomous agent frameworks, vector databases, and real-time vision systems.

Skills

Languages	**Python**, JavaScript, TypeScript, **SQL**, Java, C/C++, HTML5/CSS3
Frameworks & Libraries	**TensorFlow**, **OpenCV**, Google MediaPipe, **SentenceTransformers**, **FastAPI**, React 19, Scikit-learn, Pandas, NumPy
Tools & Databases	Git/GitHub, Docker, **ChromaDB** (Vector DB), Firebase, **PostgreSQL**, MySQL, Vercel, REST APIs

Work Experience

Tata Steel *Nov 2025 – Present*
AI/ML Intern — Chennai, India

- Engineered a **Computer Vision** pipeline using **OpenCV** for automated TMT rebar quality control, calculating rib parameters with **95% precision** — reducing defect escape rate to near zero.
- Deployed a spatial analysis module for chemical ring tests, cutting **manual inspection time by 40%** and eliminating measurement inconsistencies across 3 production lines.

Digital Info Solutions Pvt Ltd *Dec 2024 – Jan 2025*
Internship Trainee — Ghaziabad, Uttar Pradesh, India

- Built **full-stack web applications** using **Flask**, integrating Python, HTML, CSS, JavaScript, and **MySQL** to serve database-driven client interfaces.
- Delivered **3+ client projects** end-to-end, implementing RESTful backend APIs and responsive UIs that reduced client onboarding time significantly.

Projects

PlacementOS — AI Career Command Center GitHub
React 19, FastAPI, ChromaDB, SentenceTransformers, TensorFlow, Firebase

- Architected a local-first AI career platform with **9 autonomous agents**; **RAG pipeline** via **SentenceTransformers** and **ChromaDB** extracts JD metrics and generates role-tailored resumes using the Google X-Y-Z formula.
- Trained a **TensorFlow/Keras regression model** for localized salary band prediction; integrated career compass and project defense Q&A generator.

WakeGuard — AI Driver Safety Vision System GitHub
React, Vite, Google MediaPipe, Tailwind CSS, Recharts

- Tracked **478 3D facial landmarks at 60 FPS** via **MediaPipe Face Mesh**; computed **EAR**, **MAR**, and 3D head pose to classify microsleeps, yawning, and distraction with **zero backend dependencies**.
- Delivered fully **client-side inference** with voice alerts via Web Speech API — eliminating server costs entirely.

AERIS — Flood Risk Intelligence System GitHub
XGBoost, Python, FastAPI, React 19, GeoJSON

- Developed an AI flood prediction platform for Bihar delivering **7-day proactive disaster response** via **dual XGBoost models** trained on multi-variate hydraulic data from **18 river stations**.
- Integrated a **React-Leaflet GIS dashboard** with live telemetry and automated breach predictions, enabling **proactive evacuation planning**.

Education

SRM IST, Kattankulathur	B.Tech — CSE with Specialization in (AI & ML) **CGPA: 7.86 / 10**	*2023 – 2027*
A.P.J School, NOIDA	Class XII — CBSE (MPC) **68.4**%	*2023*
A.P.J School, NOIDA	Class X — CBSE (Science) **85.2**%	*2021*